

Chapter: Introduction

Ahonakpon Guy Gbaguidi

Exercise 1.11

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

By setting the derivative of the log likelihood function (1.54) with respect to μ and σ^2 equal to zero. Let restate the equation (1.54) for clarity:

$$\ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2} \ln(\sigma^2) - \frac{N}{2} \ln(2\pi) \quad (1.54)$$

In this exercise, we want to verify the results (1.55) and (1.56).

Verification of (1.55).

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n \quad (1.55)$$

Let start by computing the derivative of the log likelihood function with respect to μ :

$$\frac{\partial}{\partial \mu} \ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^N \frac{\partial}{\partial \mu} (x_n - \mu)^2 \quad (1)$$

because the other terms do not depend on μ . Now, we can compute the derivative inside the sum:

$$\frac{\partial}{\partial \mu} (x_n - \mu)^2 = -2(x_n - \mu) \quad (2)$$

Substituting this back into the expression for the derivative of the log likelihood, we get:

$$\frac{\partial}{\partial \mu} \ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (-2)(x_n - \mu) = \frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu) \quad (3)$$

We can now set this derivative equal to zero to find the maximum likelihood estimate for μ :

$$\begin{aligned}
\frac{\partial}{\partial \mu} \ln p(\mathbf{x}|\mu, \sigma^2) &= 0 \\
\frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu) &= 0 \\
\sum_{n=1}^N (x_n - \mu) &= 0 \quad (\text{since } \sigma^2 > 0) \\
\sum_{n=1}^N x_n - N\mu &= 0 \\
N\mu &= \sum_{n=1}^N x_n \\
\mu &= \frac{1}{N} \sum_{n=1}^N x_n
\end{aligned}$$

This matches the result given in (1.55).

Verification of (1.56).

$$\sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2 \quad (1.56)$$

Now, let's compute the derivative of the log likelihood function with respect to σ^2 :

$$\frac{\partial}{\partial \sigma^2} \ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2} \sum_{n=1}^N (x_n - \mu)^2 \cdot \frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \right) - \frac{N}{2} \cdot \frac{\partial}{\partial \sigma^2} \ln(\sigma^2) \quad (4)$$

We can compute the derivatives inside the expression:

$$\begin{aligned}
\frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \right) &= -\frac{1}{\sigma^4} \\
\frac{\partial}{\partial \sigma^2} \ln(\sigma^2) &= \frac{1}{\sigma^2}
\end{aligned}$$

Substituting these back into the expression for the derivative of the log likelihood, we get:

$$\frac{\partial}{\partial \sigma^2} \ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2} \sum_{n=1}^N (x_n - \mu)^2 \cdot \left(-\frac{1}{\sigma^4} \right) - \frac{N}{2} \cdot \frac{1}{\sigma^2} \quad (5)$$

Simplifying this expression:

$$\frac{\partial}{\partial \sigma^2} \ln p(\mathbf{x}|\mu, \sigma^2) = \frac{1}{2\sigma^4} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2\sigma^2} \quad (6)$$

We can now set this derivative equal to zero to find the maximum likelihood estimate for σ^2 :

$$\begin{aligned}
\frac{\partial}{\partial \sigma^2} \ln p(\mathbf{x}|\mu, \sigma^2) &= 0 \\
\frac{1}{2\sigma^4} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2\sigma^2} &= 0 \\
\sum_{n=1}^N (x_n - \mu)^2 &= N\sigma^2 \\
\sigma^2 &= \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2
\end{aligned}$$

This matches the result given in (1.56).

We can now conclude that the maximum likelihood estimates for μ and σ^2 are given by (1.55) and (1.56) respectively, and both estimates are unique because the log likelihood function is concave with respect to μ and σ^2 .